

Leveraging environmental variability in dissolved oxygen on tropical reefs to predict coral ecology in a changing climate

Background and Rationale: Tropical coral reefs around the world are threatened by a combination of intensifying anthropogenic factors, which have triggered drastic declines in reef-building coral abundance and diversity through mass bleaching (i.e., the loss of photosynthetic symbionts from the coral host animal)¹ and habitat degradation². Increasing temperature has been considered the primary driver of coral declines³, but a growing body of evidence suggests low oxygen (O₂) availability, alone and in combination with increased temperature, can trigger mortality^{4,5,6}. Similarly, reduced O₂ degrades habitat quality by limiting coral energy production, favoring macroalgal-dominated communities, and limiting recruitment of offspring^{5,7}.

Decreases in O₂ availability, i.e., deoxygenation, are expected to intensify with climate change⁸, prompting recent efforts to classify aquatic deoxygenation as the next planetary boundary⁹. Despite this, environmental stress on natural reefs is most commonly quantified with tools based on temperature alone¹⁰, e.g., Degree Heating Weeks (DHW) for cumulative heat stress and Maximum Monthly Mean (MMM) for historical thermal conditions¹¹. **Therefore, this project will develop standardized metrics of reef oxygen dynamics, test their power to predict bleaching events, and explore how oxygen environments shape coral ecology and symbiosis.** This characterization of reef O₂ dynamics and low O₂ stress will be vital for reef restoration practitioners, environmental policy makers, and stakeholders to establish historical baselines, track ongoing stress events, and predict outcomes of intensifying deoxygenation as climates change¹².

Descriptive and synoptic tools to characterize reef dissolved oxygen (DO) dynamics and low O₂ stress will require consideration of the unique scales at which changes in O₂ occurs. Reef O₂ availability is controlled by climatic, biological, and chemical factors which, along with the natural diel cycling of DO, can create acute high intensity low O₂ events (hours to days), periods of moderate deoxygenation interrupted by daily reoxygenation (weeks to seasons), and long-term downward shifts in average O₂ conditions (years to decades)^{5,7}. To date, the spatiotemporal variation in O₂ conditions on reefs has been described without enough nuance to effectively incorporate O₂ environments into assessments of ongoing stress events, mass coral mortality, and degradation of reef habitats. **To fill this gap, the research proposed here will develop innovative tools to filter inherently noisy reef DO data to capture trends, variability, and predictability over the time scales in which deoxygenation operates.** Similar to DHW and MMM, these O₂-specific descriptors will allow reef stakeholders to leverage DO to more accurately characterize and predict the status of their reefs.

Although all corals require O₂ for their predominant energy production pathways, the ability to resist and tolerate hypoxia caused by deoxygenation differs between species¹³. Therefore, intensifying deoxygenation is expected to not only reduce overall coral abundance, via bleaching or habitat degradation, but also shift reef communities, structure, and function⁶. Similarly, the communities of coral algal symbionts (Family Symbiodiniaceae) living *in hospite*¹⁴ and free-living¹⁵ vary with environmental conditions. However, whether DO contributes to changes in host-symbiont dynamics and niche partitioning remains unclear. Considering the global scales at which deoxygenation operates^{5,7,8}, ignoring O₂ conditions will leave marine management policies, emergency rescue activities, resource allocation, and reef recovery planning in the dark. Therefore, prioritizing tools to quantify reef O₂ environments will allow this universal threat to be incorporated into the coral reef crisis response.

Research Questions:

- (Q1) What are the relevant components of O₂ environments that can characterize reefs across space, time, and/or during stress events?*
- (Q2) What signals of low O₂ stress help explain variation in mass bleaching events?*
- (Q3) How does the O₂ environment contribute to ecological, physiological, and geographic variation observed in coral hosts and their algal symbionts?*

Approach: Reef DO data have been archived by government programs, philanthropic organizations or in association with scientific publications (see Data Description) resulting in a patchwork of DO

time series of variable length and resolution. However, in the absence of satellite-derived data capable of representing benthic DO concentration, pooling data sources is not only practical, but it offers an opportunity to operate within the constraints many reef stakeholders already experience while balancing the costs of collecting DO data and the desire to track reef environments.

Q1: I will produce a set of candidate descriptive features of reef O₂ dynamics, beginning with DO metrics identified in other fields, e.g., daily/interannual variability¹⁶ and autocorrelation¹⁷, and then employing automated time series feature detection to explore aspects of *in situ* O₂ environments at increasing complexity. Candidate features will then be evaluated using statistical tests to maximize the potential for identifying differences between reef sites, seasons, years, and/or anomalies. Features will be tested across datasets with decreasing sampling duration and frequency to evaluate their applicability in limited DO data scenarios.

Q2: To test the power of reef O₂ environments to predict coral bleaching, candidate features will be used to characterize conditions on reefs prior to, during, and following the mass coral bleaching events in Australia, Indo-Pacific, and Caribbean that have been documented *in situ* by coral monitoring programs (see Data Description). Previous stress exposure can harden corals to future stress¹⁸, so I will extend reef O₂ environment characterization back months or years prior to bleaching events (when possible) to uncover historical predictors of bleaching resilience. Metrics quantifying the accumulation of O₂ stress throughout bleaching events will provide indicators of reef-wide bleaching susceptibility. To test this, I will use Bayesian generalized linear modeling to assess the impact of O₂ features on the presence and severity of bleaching, both alone and in conjunction with the most common temperature-based bleaching predictors.

Q3: Reef monitoring programs, annotated satellite imagery, genetic data repositories (see Data Description) provide opportunities to track changes in coral cover and species distributions over space and time in relation to O₂ conditions. To achieve this, I will first extract ecological metrics from sites with DO data. Coral cover and species presence, abundance, and diversity serve as indicators of reef health (increased cover and abundance) and community state (species relative abundances) which will be correlated with O₂ features in a Bayesian GLM framework. Photosymbiont community composition will be assessed with ITS2 genetic markers mined from symbiont genetic and transcriptomic databases, allowing taxa ranges and coral-algae partnership dynamics to be mapped in relation to O₂ environments. I will test the effect of O₂ environments on coral and algae species distributions using MaxEnt Species Distribution Models with O₂ features as predictors. Given the global distribution of focal reefs, I will also assess the relative strength of O₂ features on coral ecology among regions to identify “hotspots” at greater threat of intensifying deoxygenation.

Relevance to Previous Experience and ESIIL’s Mission: The proposed project builds on my expertise in coral deoxygenation ecology which includes both mapping the environmental and physiological landscape of low O₂ in corals. I have applied many of the technical skills required in the proposed research through my work in coral and algal symbiont ecological genomics^{19,20}, environmental characterization²¹, and open science²². Similarly, my work in applied settings^{20,21,22} has enabled me to ground basic science with practicality to improve the development and accessibility of new reef tools. The proposed work aligns with ESIIL’s mission by leveraging open-access data from disparate archives to simultaneously explore an emerging planetary boundary and generate applied tools for addressing the ongoing coral crisis. I am excited to contribute to the interdisciplinary environment within ESIIL and expand the proposed work by learning from systems which have tackled similar problems, while also using my unique background to help others.

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